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April 8, 2022

Dr. Stephanie Johnson
Office of Energy Efficiency and Renewable Energy
Building Technologies Office
Department of Energy
1000 Independence Avenue, SW
Washington, DC 20585-0121

Docket Number: EERE-2021-BT-STD-0035 and EERE-2021-BT-TP-0036
RIN: 1904-AF26

Dear Dr. Johnson:

This letter comprises the comments of the Pacific Gas and Electric Company (PG&E), San Diego Gas and Electric (SDG&E), and Southern California Edison (SCE) in response to the United States (U.S.) Department of Energy (DOE) Request for Information (RFI) on Air Cleaner Standards and Test Procedures.

The signatories of this letter, collectively referred to herein as the California Investor-Owned Utilities (CA IOUs), represent some of the largest utility companies in the Western U.S., serving over 32 million customers. As energy companies, we understand the potential of appliance efficiency standards to cut costs and reduce consumption while maintaining or increasing consumer utility of products. We have a responsibility to our customers to advocate for standards that accurately reflect the climate and conditions of our respective service areas.

We appreciate this opportunity to provide the following comments about this RFI. As noted in our response to the Notice of Proposed Determination (NOPD), we agree with DOE's tentative determination that coverage of air cleaners is necessary and appropriate to carry out the purposes of the Energy Policy and Conservation Act of 1975 (EPCA) and that the criteria to prescribe energy conservation standards are met. We recommend an integrated active and standby mode performance metric based on a PM 2.5 Clean Air Delivery Rate (CADR) performance metric and ask DOE to account for the benefits of design options that reduce operating hours and fan speed without compromising performance. We fully support DOE's consideration of this product category and offer the following comments:

- 1. The CA IOUs are currently engaged in productive discussions with several stakeholders that are representative of relevant points of view to help advance the development of this and subsequent air cleaner rulemakings. While these talks are ongoing, we ask DOE to consider any recommendations that develop from these discussions.**

A diverse stakeholder working group is currently developing recommendations regarding definitions, scope, performance metrics, test procedures, and efficiency standards. In addition to

the topics put forth in response to the NOPD,¹ there is general alignment on definition, scope, and tested particles.

Information below highlight key areas the CA IOUs believe warrant a specific mention at this rulemaking stage but do not necessarily supersede any recommendations that eventually come from this working group.

2. The CA IOUs reiterate our comments from the NOPD that the criteria to prescribe energy conservation standards are met.

As noted in our response to the September 2021 NOPD,² we agree with DOE’s tentative determination that coverage of air cleaners is necessary and appropriate and that the conditions to prescribe energy conservation standards have been met. The air cleaner market has been growing for years, and this growth has accelerated amid the COVID-19 pandemic and increased prevalence of wildfires in California and the Western United States. Market information provided in our November 2021 NOPD comment letter provides publicly available data that demonstrates the market size and energy usage requirements to prescribe energy conservation standards per EPCA have been met for air cleaners.

Additionally, sales data from PG&E’s participation in the ENERGY STAR® Retail Products Platform (RPP) program shows that portable air purifiers sales in California are dominated by products that achieve either ENERGY STAR Version 1.0 or 2.0. However, this market share dominance appears to be on a downward trend, as the data in Figure 1 indicates. Data procured by the CA IOUs from the NPD Group indicate that the percentage of ENERGY STAR products sold in California closely mirrors the United States as a whole.³ EPA estimated that products meeting ENERGY STAR Version 2.0 use an average of 300 kWh/yr and that non-ENERGY STAR units use an estimated 450 kWh/year. This supports the fact that air cleaners exceed the significant energy use thresholds prescribed in EPCA.⁴

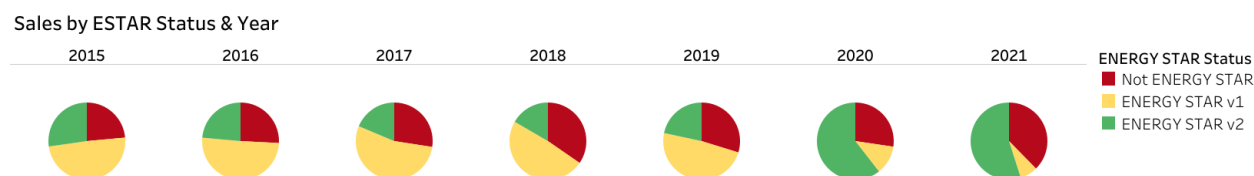


Figure 1: ENERGY STAR Market Saturation Based on RPP Data

Source: CA IOU analysis of the PG&E ENERGY STAR Retail Product Platform sales data.⁵

3. A performance metric based on PM 2.5 CADR for air cleaners is appropriate; however, we encourage DOE to track product performance of pollen and microbials to ensure PM 2.5 remains an appropriate representation of an average use cycle.

¹ <https://www.regulations.gov/comment/EERE-2021-BT-DET-0022-0012>

² <https://www.regulations.gov/comment/EERE-2021-BT-DET-0022-0008>.

³ See Table 4 of the Air Purifier NPD Analysis memo submission for supporting data. A reference to this memo and accompanying data is discussed in comment 7.

⁴ Air Purifiers (Cleaners) | Products | ENERGY STAR: https://www.energystar.gov/products/air_purifiers_cleaners and https://www.epa.gov/sites/default/files/2018-07/documents/residential_air_cleaners_-_a_technical_summary_3rd_edition.pdf

⁵ Approximately 30 percent of all RPP sales could be matched with a model number, a requirement to ensure it met ENERGY STAR version 1.0 of 2.0. These models were listed as “not ENERGY STAR” by default.

On behalf of the CA IOUs, Evergreen Economics conducted a nationally representative survey of air purifier owners and found that the top four most popular reasons for purchasing an air purifier is reducing allergy symptoms, dust, pollen, and fine particulate matter. Each of these four factors were deemed to be at least “very important” motivators by 60 percent or more of respondents as seen in Figure 2.

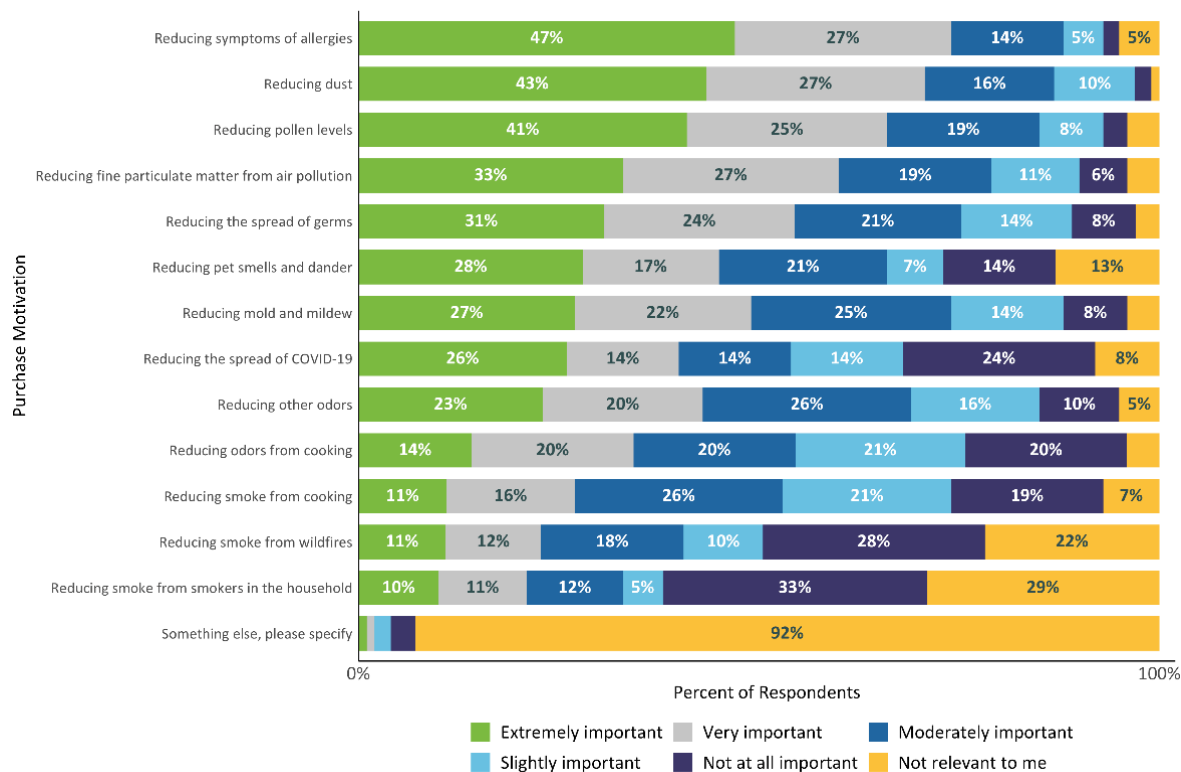


Figure 2: Air cleaner purchasing motivations (n=491)

Source: Evergreen Economics Consumer Air Purifier Study Results.

The PM 2.5 CADR metric is represented by a geometric average of cigarette smoke and dust CADRs. The CA IOUs performed an analysis of the ENERGY STAR qualified product list (QPL) to see how the rank order of products would be impacted when comparing a metric based on PM 2.5 CADR and an average of the three ENERGY STAR tested particles, cigarette smoke, dust, and pollen; the results are shown in Table 1. An average of the three ENERGY STAR tested particles was chosen as a point of comparison to PM 2.5, as all three are in the top four of consumer air cleaner purchase motivations. We found that there is some degree of change in rank order when comparing these two blended CADRs, with an average rank order change of six places when looking at all ENERGY STAR models. However, 54 percent of models have a rank order change of four or less. The rank order changes are less pronounced when looking at the top-performing models. Therefore, if a consumer desires a top-performing model that is based on a PM 2.5 performance metric, they are likely to receive a unit that performs well in pollen as well. Pollen is categorized under two of the top three consumer purchase motivations.

Table 1: Analysis of the ranking change for air cleaner models in the ENERGY STAR QPL

CADR Relative Rank Order Position Change	All		Top 100 (PM 2.5)		Top 25 (PM 2.5)		Bottom 100 (PM 2.5)		Bottom 25 (PM 2.5)	
	Count	Spread	Count	Spread	Count	Spread	Count	Spread	Count	Spread
0	24	8%	9	9%	6	24%	7	7%	2	8%
1	48	17%	25	25%	8	32%	13	13%	7	28%
2	30	11%	12	12%	3	12%	10	10%	4	16%
3	26	9%	12	12%	7	28%	4	4%	2	8%
4	26	9%	10	10%	0	0%	8	8%	5	20%
5	14	5%	5	5%	0	0%	5	5%	2	8%
6	14	5%	2	2%	0	0%	3	3%	0	0%
7	13	5%	3	3%	1	4%	6	6%	0	0%
8	10	4%	1	1%	0	0%	3	3%	0	0%
9	11	4%	0	0%	0	0%	5	5%	0	0%
10	6	2%	1	1%	0	0%	5	5%	0	0%
11	4	1%	2	2%	0	0%	1	1%	0	0%
12	8	3%	4	4%	0	0%	4	4%	2	8%
13	5	2%	0	0%	0	0%	3	3%	0	0%
14	12	4%	9	9%	0	0%	2	2%	0	0%
15	4	1%	2	2%	0	0%	1	1%	0	0%
16	5	2%	1	1%	0	0%	3	3%	0	0%
17	1	0%	0	0%	0	0%	1	1%	0	0%
18	3	1%	0	0%	0	0%	1	1%	0	0%
19	2	1%	1	1%	0	0%	0	0%	0	0%
20	3	1%	0	0%	0	0%	3	3%	0	0%
21	2	1%	0	0%	0	0%	1	1%	1	4%
22	0	0%	0	0%	0	0%	0	0%	0	0%
23	2	1%	0	0%	0	0%	2	2%	0	0%
24	7	2%	0	0%	0	0%	7	7%	0	0%
25	0	0%	0	0%	0	0%	0	0%	0	0%
26	0	0%	0	0%	0	0%	0	0%	0	0%
27	0	0%	0	0%	0	0%	0	0%	0	0%
28	2	1%	0	0%	0	0%	2	2%	0	0%
29	0	0%	0	0%	0	0%	0	0%	0	0%
30	1	0%	1	1%	0	0%	0	0%	0	0%
Avg Position Change	6		5		2		9		4	
Number of models	283		100		25		100		25	

Source: CA IOU analysis of ENERGY STAR QPL; Exported on March 11th, 2022.

Additionally, the CA IOUs understand that there are low air-flow scope concerns for active mode testing with pollen that do not exist for cigarette smoke and dust. AHAM AC-1 indicates that testing with pollen particles is not considered sufficiently accurate (based on a 95 percent confidence limit) for products below 25 cubic feet per minute (CFM) and thus products of that size are out of scope for AHAM AC-1. On the other hand, cigarette smoke and dust particles can be considered sufficiently accurate down to a CADR of 10 CFM. This level of confidence translates the expected precision of the AC-1 active mode test, with pollen listed at ± 25 CFM CADR, while cigarette smoke and dust are both listed at ± 10 CFM CADR. To ensure products with low CFMs can be included in the scope and to produce the most precise test procedure that balances the representativeness of consumer use cases, the CA IOUs recommend that DOE adopt a performance metric based on PM 2.5 CADR, but we encourage DOE to monitor pollen CADR performance to ensure a strong correlation is maintained between PM 2.5 and pollen performance. If this relationship were to significantly degrade, a reconsideration of the performance metric would be justified.

Given the increase in purchases of air cleaners during the COVID-19 pandemic, as shown in Figure 3,⁶ we ask that DOE also work with the relevant industry participants to develop testing requirements for microbial agents (e.g., bacteria, mold, and viruses). The same Evergreen Economic survey found that 55 percent of respondents listed reducing the spread of germs as either “very important” or “extremely important” in motivating their purchase of air cleaners.

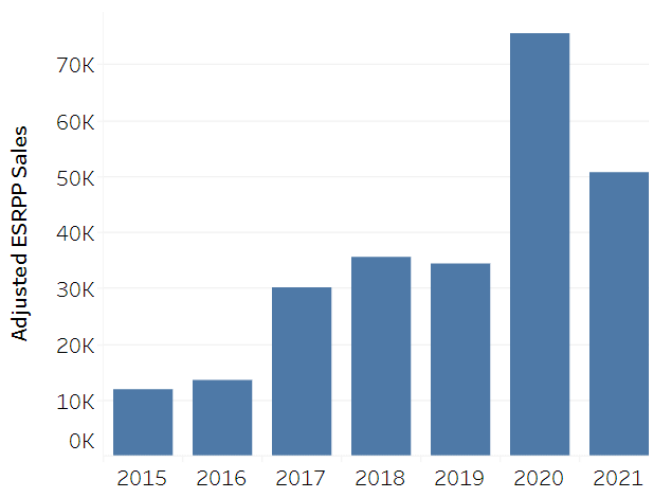


Figure 3: ENERGY STAR Retail Product Portfolio (ESRPP) sales of air cleaners for 2015 through 2021

Source: CA IOU analysis of the PG&E ENERGY STAR Retail Product Platform sales data.

⁶ Analyzing the RPP sales data, there was a more than doubling of sales in 2020, likely in response to the COVID 19 pandemic. Reflected in Figure 3, years 2016, 2017, 2018, and 2021 have full-year data. 2015 is missing data from January and February, 2019 is missing data from Q2 - Q4, and 2020 is missing data from Q1. 2015 - 2019 only include in-store sales, while 2020 - 2021 also include online sales. Online sales made up roughly half of the sales in 2020 - 2021. Additionally, the 2020 - 2021 data only includes sales from Home Depot, Lowe's, and BestBuy, while the prior years include sales from those retailers and Sear's. For years where there were not full-year sales, this chart multiplies the raw sales by a factor to account for missing months. For years where online sales were not included, this chart multiplies the raw sales by a factor to account for missing online sales. The factors for missing months are empirically derived from the distribution in years with full-year data, and the factor for missing online sales is empirically derived from the years which include online sales. Since the years with online sales were post-pandemic, it is possible that the factor to account for missing online sales overestimates the online sales from 2015 - 2019.

Two general approaches are possible for handling microbials: filtration and deactivation. Complications include but are not limited to:

- a) Microbials are often associated with particles;
- b) Health impacts vary with quantity ingested and among different biological entities;
- c) The degree of lung penetration may favor smaller particles (<2.5 microns);
- d) Larger particles fall out of the air sooner, while smaller particles may persist; and
- e) Environmental factors (e.g., humidity, temperature, sunlight, and cross-currents) can influence the persistence of particles in the air and the viability of biological agents.

We understand that microbials are often carried and transmitted by particulates; thus, a correlation in performance between CADR for dust, pollen, and cigarette smoke and biological agents is to be expected. In the absence of an acceptable standardized energy performance rating for biological agents, the CA IOUs find it reasonable to focus on the accepted particulate-based energy testing standard but encourage DOE to monitor biological testing in this context and validate if such a correlation exists.

4. The CA IOUs urge DOE to account for the energy benefits of technologies that reduce either fan speed and/or operating hours while maintaining air cleanliness.

The vast majority of air purifiers include a fan, whether the air cleaning is performed by HEPA, carbon, or electrostatic filters; includes ultraviolet light; or generates ions or ozone. The fan is also the main energy-consuming component in most air purifiers.

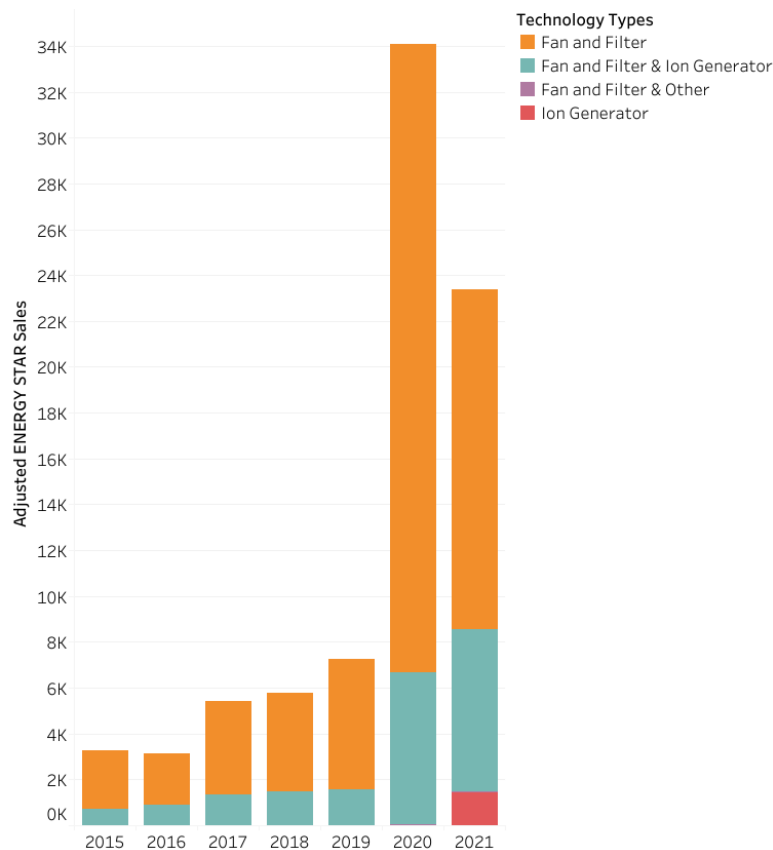


Figure 4: Technologies used in Air Cleaners by Sales

Source: CA IOU analysis of the PG&E ENERGY STAR Retail Product Platform sales data.

This is especially true with air cleaners that use filters, which represent more than 70 percent of the air cleaners in the Evergreen Economics national survey, as seen in Figure 5 (HEPA filtered, carbon filtered, and electrostatic). Therefore, the impact of design options that reduce average fan speed and/or operating hours at high speeds without compromising overall air cleanliness should be appropriately evaluated.

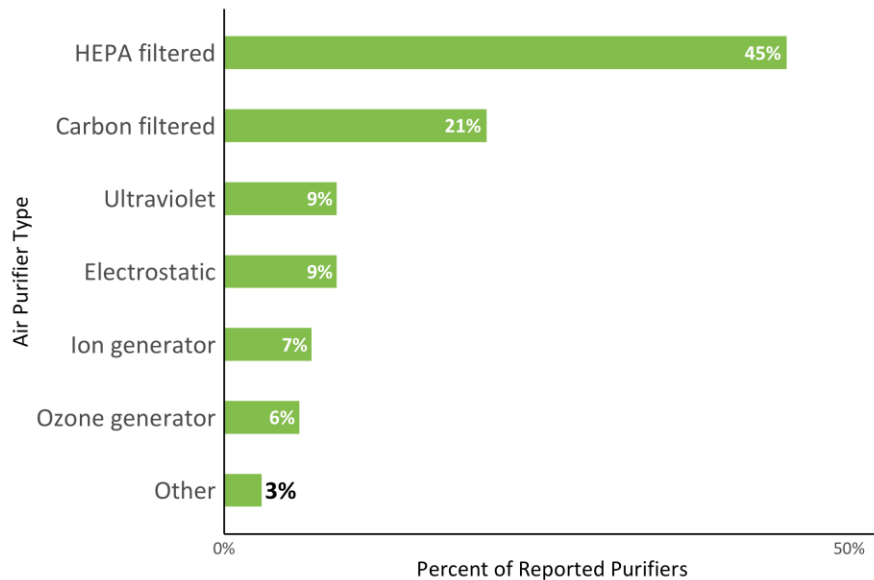


Figure 5: Air purifier type owned by respondents (n=764)

Source: Evergreen Economics Consumer Air Purifier Study Results.

Fan speed reduction can be accomplished by a combination of variable speed fans and the necessary sensors and controls to detect improvements in air quality and reduce fan speed when the air is deemed adequately clean. This is not quantified in the current AHAM AC-1 test procedure, which only tests operation at maximum speed for a short duration. Additionally, for products with fans and other energy-consuming components, sensors and controls that turn down or turn off the product when the air quality is sufficiently improved are likely to provide significant energy savings. The Evergreen Survey asked respondents whether their air purifiers had these types of automatic settings and found all technology types had a significant percentage with automatic settings, ranging from 28 and 29 percent for ion generators and HEPA filters, respectively, all the way up to 45 percent for electrostatic products, as seen in Figure 6. The prevalence of this design option makes it very important to quantify the energy benefits.

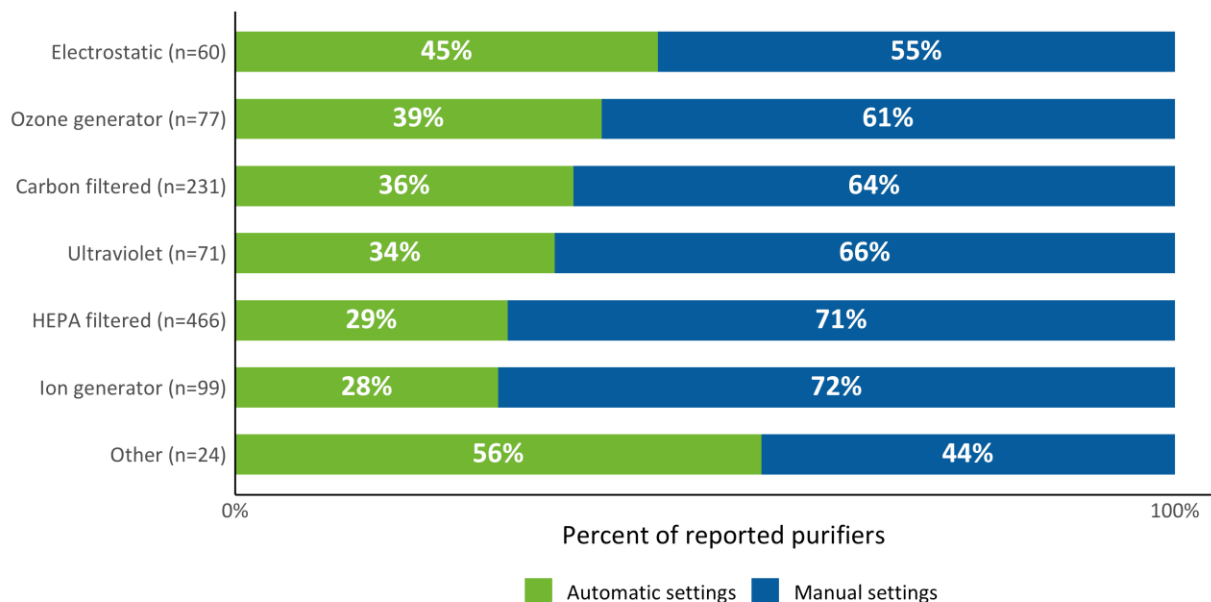


Figure 6: Incidence of air cleaners with automatic settings

Source: Evergreen Economics Consumer Air Purifier Study Results.

This significant market penetration of units with automatic controls is supported by sales data procured by the CA IOUs, indicating that of approximately 4 million portable air purifiers sold in the United States between November 2020 and October 2021, or 35 percent, had automatic controls.⁷ We ask that DOE investigate how to quantify the energy efficiency that is achieved by air quality sensors and controls in air cleaner products and what conditions (e.g., if the automatic setting is the as-shipped default operation) would be appropriate to test an automatic setting.

5. An integrated performance metric that appropriately allocates active, standby, and off mode operating hours should be implemented for air cleaners.

It is technically feasible to integrate measures of standby and off mode energy consumption into an overall performance metric for air cleaners. Some standby operations aid in reducing active mode operating hours, such as feedback loop technologies that monitor air quality to determine when air purification is needed. These technologies may use more standby power but can result in lower energy consumption on a per annum basis.

The survey conducted by Evergreen Economics found that there is a large range of active mode operating hours, with 23 percent of respondents reporting that they run their air cleaners all day, but 53 percent saying that they run their air purifier for 6 hours or fewer, as seen in Figure 7. We ask that DOE review the survey information when considering time spent in each mode and to consider the prevalence of automatic settings when deciding on appropriate weighting factors. DOE should test standby power in the as-shipped condition, with any manufacturer's recommended settings for normal use enabled.

⁷ See Table 4 of the Air Purifier NPD Analysis memo submission for supporting data. A reference to this memo and accompanying data is discussed in comment 7.

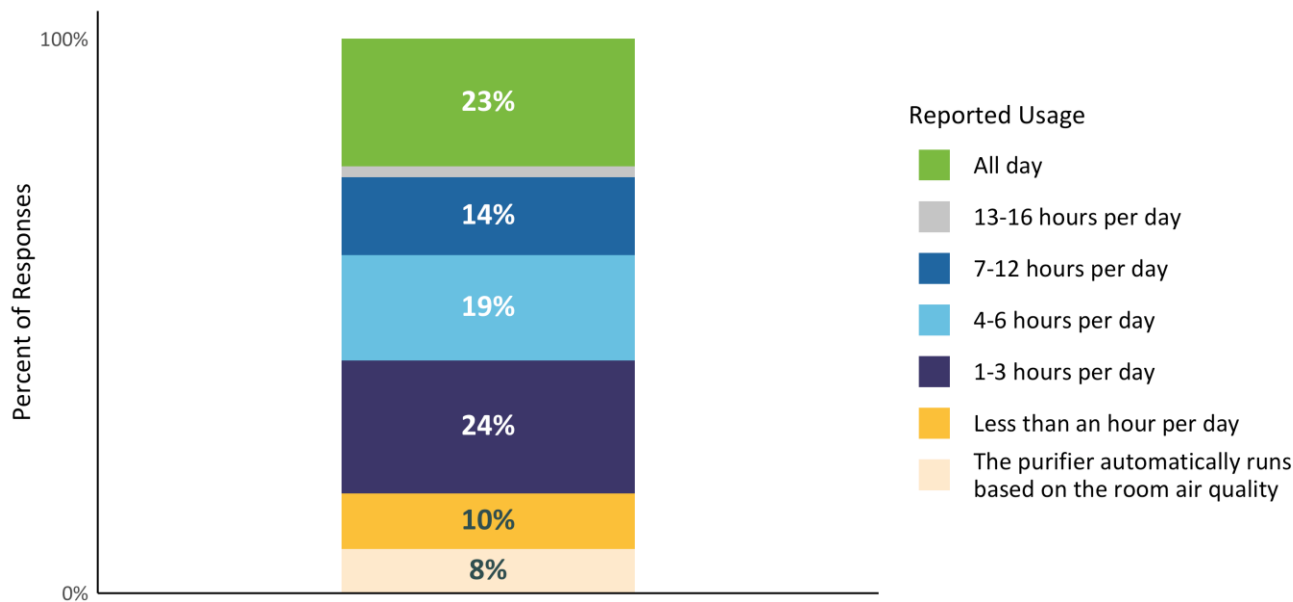


Figure 7: Reported usage per day (n=488).

Source: Evergreen Economics Consumer Air Purifier Study Results.

6. Mounted and whole-home/in-duct units should be within the scope of this rulemaking, and we recommend that DOE also include UV lamps that are marketed as air cleaners.

The CA IOUs support DOE’s proposal in the NOPD to cover a more comprehensive range of the consumer market for air cleaning and purification, rather than just portable air cleaners. Since the definition proposed by DOE specifies that air cleaners need to be powered, it is clear that these must be energy-consuming products and not unpowered mechanical or electrostatic air filters. One example of a product that we believe would be in scope based on the proposed definition is a whole home air purifier designed for use with a forced air indoor air moving system, such that the whole home air purifier is powered by the terminals of the air moving system.⁸ While other DOE test procedures and energy conservation standards account for the energy consumption of such air moving systems (e.g., consumer furnaces), the filter provisions in those test procedures do not currently take into account the energy consumption of such a whole home air purifier reliant on the forced air system’s terminals for the power supply. Such an air cleaner is likely to consume electrical energy nonstop for the life of the air cleaner. Whole home electronic air cleaners with an air-proving switch can automatically cycle the air cleaner on or off with the system fan.⁹ In such instances, the energy consumption of the whole home electronic air cleaner would be a function of the run times associated with the HVAC system’s furnace fan, modular blower, or air handler. The run times would also vary depending on the indoor blower type (i.e., a constant-air-volume-rate indoor blower, a variable-speed constant-air-volume-rate indoor blower, or a variable-speed, variable-air-volume-rate indoor blower).

⁸ Technical specifications can be accessed here (page 2 of each document confirms typical input power of 18 Watts or maximum power consumption of 25 Watts): <https://www.americancoolingandheating.com/wp-content/uploads/carrier-product-data-specifications/gapaa-3pd.pdf>; https://resource.carrierenterprise.com/is/content/Watscocom/carrier_dgapaxx_ss

⁹ <https://www.cleancleancomfort.com/pdfviewer.aspx?pdfurl=docs/librariesprovider10/default-document-library/ss-ae14.pdf?view=true>

A second example of a whole-home ducted air cleaner that we believe would be in scope based on the proposed definition is the Clean Comfort CC-HEPA, which can be installed in a bypass configuration, has its own plug-in power cord (120 VAC) and its own integrated ventilator.¹⁰ A product of this type should be covered by DOE as an air purifier. The energy consumption of the integrated ventilator could potentially be addressed via a CFM/W metric rated in accordance with Home Ventilation Institute's (HVI) Standard 916 Air Flow Test Procedure. The current AHAM test procedure for portable air cleaners could be modified to arrive at a CADR/W metric for these products.¹¹ Powered ventilators¹² exist today in the market that are certified by HVI¹³ for rated air flow and input power. While powered ventilators play a role in maintaining a comfortable indoor environment, manufacturers in some cases also claim that such products help reduce airborne particulates from the incoming air stream.¹⁴ Both energy efficiency and efficacy considerations are key for such products.

Additionally, the CA IOUs recommend that DOE modify the proposed definition to include air cleaning products that function only by means of ultraviolet (UV) light to cover a more comprehensive range of the consumer market. The *ANSI/ASHRAE Standard 185.1-2020*¹⁵ and *ANSI/ASHRAE Standard 185.2-2020*¹⁶ provide methods for determining the efficacy of UV air cleaning products to inactivate airborne microorganisms and for determining the intensity of the lamps, respectively. A third example for DOE's consideration is a duct-mounted air purifier with UV light and a turbulator.¹⁷ UV light is being increasingly used in efforts to inactivate microbes.¹⁸ ASHRAE's initial focus on ventilation in response to the COVID-19 pandemic has been shifting toward germicidal UV (GUV) because of the energy benefits (e.g., in one example, GUV lamps achieved 73-80 percent effectiveness against airborne pathogens while reducing equivalent ventilation requirements four-fold from 24 air changes per hour to six). However, current GUV lamps, including LEDs, are very inefficient (<1% to 10%); in particular, heat shedding has been a major issue in developing new GUV lamps.¹⁹

Of note, DOE's definition for general service fluorescent lamps excludes "lamps primarily designed to produce radiation in the ultra-violet region of the spectrum,"²⁰ and therefore fluorescent UV lamps would not be covered under the lamp standard.²¹ It is unclear if UV air cleaners relying on other UV lamp technologies could be covered under DOE's pending General Service Lamp (GSL) regulations. Based on our review, these products do not provide general illumination and would therefore not be covered as a GSL.²² In both cases, the CA IOUs

¹⁰ <https://www.cleancomfort.com/pdfviewer.aspx?pdfurl=docs/librariesprovider10/default-document-library/ss-cc-hepa.pdf?view=true>.

¹¹ AHRI Standard 680-2017 and AHRI Standard 850-2013 prescribe reporting requirements for input power but appear to exclude a test method that is representative of an average use cycle of an air cleaner for which electrical energy consumption is relevant. Electronic air cleaners are in scope for both AHRI standards.

¹² <https://daikincomfort.com/daikin-one-powered-ventilator>

¹³ Select "Daikin" as brand name in the following hyperlink: <https://www.hvi.org/hvi-certified-products-directory/section-i-complete-product-listing/>

¹⁴ See "Key Features and Benefits" section [here](#).

¹⁵ Method of Testing UVC Lights for Use in Air Handling Units or Air Ducts to Inactivate Airborne Microorganisms.

¹⁶ Method of Testing Ultraviolet Lamps for Use in HVAC&R Units or Air Ducts to Inactivate Microorganisms on Irradiated Surfaces.

¹⁷ <https://www.cleancomfort.com/pdfviewer.aspx?pdfurl=docs/librariesprovider10/default-document-library/ss-ua-cb.pdf?view=true>

¹⁸ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7571309/>

¹⁹ Illuminating Engineering Society webinar, February 17 2022, Introduction to RP-44-21, Ultraviolet Germicidal Irradiation (UVGI) <https://www.ies.org/event/webinar-introduction-to-rp-44-on-germicidal-ultraviolet/>

²⁰ 10 CFR 430.2.

²¹ <https://www.govinfo.gov/content/pkg/CFR-2013-title10-vol3/pdf/CFR-2013-title10-vol3-sec430-2.pdf>

²² <https://www.regulations.gov/document/EERE-2021-BT-STD-0005-0034>

recommend that products marketed for air cleaning be regulated as air cleaners, in line with their intended use.

The CA IOUs recommend DOE also consider provisions in other ANSI/ASHRAE standards that may directly impact products under the scope of this rulemaking. For example, ANSI/ASHRAE Standard 62.1-2019 prescribes a UL 2998-specific listing and labeling requirement in Section 5.7.1 for air-cleaning devices potentially generating ozone. Section 5.7.2 limits UV generating devices to 185 nm in wavelength.

The CA IOUs recognize that whole-home air purification solutions raise a complexity of issues pertaining to the applicability of test procedures based on the installed application. We have provided DOE with multiple examples where the energy use of such air purification systems is not addressed in DOE's current test procedures for consumer products and references to existing test procedures and building standards provisions that may serve DOE in its efforts to establish appropriate test procedures for such whole-home air purification systems.

7. The CA IOUs have docketed the full Evergreen Economics survey memo and NPD sales data memo for DOE's consideration.

The data in these memos should provide DOE with helpful information as they complete the test procedure and standard rulemakings for air cleaners. The information in these memos will be particularly useful in answering questions regarding consumer usage of air cleaners, potential impacts of COVID-19 on the air cleaner market, annual operating hours, market drivers, consumer expectations, and market characteristics related to efficiency and feature distribution. The national consumer survey was conducted in late 2021 and reflects very recent market information, as influenced by the COVID-19 pandemic and recent increases in wildfires. This memo also provides details on hours of use beyond those cited in CA IOUs' November 2021 NOPD comments. The NPD sales data analysis memo is based on portable air cleaner sales from October 2020 to October 2021.

In conclusion, we would like to reiterate our support on DOE's test procedure and standards RFI for air cleaners. We thank DOE for the opportunity to be involved in this process.

Sincerely,



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